

VANAVIL ARTS AND SCIENCE COLLEGE

DEPARTMENT OF COMPUTER SCIENCE

BACHELOR OF SCIENCE

B.Sc. COMPUTER SCIENCE

A PROJECT REPORT ON

"COURIER MANAGEMENT SYSTEM"

A Salesforce-Based Application for Efficient Courier Management

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****SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS**

FOR THE AWARD OF THE DEGREE OF
B.Sc. COMPUTER SCIENCE**

UNDER THE GUIDANCE OF

Department of Computer Science

ACADEMIC YEAR

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COURIER MANAGEMENT SYSTEM

1. Project Overview

The Courier Management System is a Salesforce-based application used to manage the complete courier delivery process. It helps to store customer details, courier booking details, parcel information, delivery agent details, payment details, and delivery status in one centralized system.

The system makes courier management easier, faster, and more organized.

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2. Problem Statement

Managing courier details manually can be difficult and time-consuming. It may lead to incorrect records, delayed updates, difficulty in tracking parcels, and problems in maintaining customer information.

The Courier Management System solves these problems by providing a centralized Salesforce application for managing courier operations.

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3. Objectives

- **To manage customer details.**
- **To create and manage courier bookings.**
- **To maintain parcel information.**

- **To assign delivery agents.**
- **To track courier delivery status.**
- **To manage payment details.**
- **To reduce manual work.**
- **To generate reports.**
- **To provide a dashboard for courier management.**

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4. Main Modules

Customer Management

Stores customer name, phone number, email, and address.

Courier Booking

Used to create a new courier booking and store parcel details.

Parcel Management

Maintains parcel type, weight, source, destination, and delivery date.

Delivery Management

Assigns couriers to delivery agents and manages delivery status.

Payment Management

Stores courier charges, payment method, and payment status.

Courier Tracking

Tracks the courier from booking to final delivery.

Reports and Dashboard

Displays courier statistics and delivery information.

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5. Courier Delivery Process

Customer

↓

Courier Booking

↓

Parcel Pickup

↓

Processing

↓

In Transit

↓

Out for Delivery

↓

Delivered

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6. Courier Status

The system can use the following statuses:

- **Booked**
- **Picked Up**
- **In Transit**
- **Out for Delivery**
- **Delivered**
- **Cancelled**
- **Returned**

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7. Salesforce Objects

Object	Purpose
Customer	Stores customer information
Courier	Stores courier booking details
Parcel	Stores parcel information
Delivery Agent	Stores delivery agent details
Payment	Stores payment information

8. Important Courier Fields

Courier Object

- **Courier ID**
- **Customer Name**

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  Sender Name

  Receiver Name

- **Source**
- **Destination**
- **Parcel Type**
- **Parcel Weight**
- **Booking Date**
- **Expected Delivery Date**
- **Delivery Status** • **Delivery Agent**
- **Courier Charge**

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9. Automation

Salesforce Flow can be used to automate courier management.

For example:

Courier Created → Status automatically becomes "Booked" When the

parcel is delivered:

Status = Delivered

The system can automatically update the delivery date and send a notification.

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10. Reports

The system can generate:

- **Total Courier Report**
- **Delivered Courier Report**
- **Pending Courier Report**
- **In-Transit Courier Report**
- **Cancelled Courier Report**
- **Payment Report**

  Delivery Agent Report

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Monthly Courier Report

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11. Dashboard

The Salesforce dashboard can display:

Total Couriers | Delivered | In Transit | Pending | Cancelled It

can also show charts for:

- **Daily bookings**
- **Monthly deliveries**
- **Courier status**
- **Agent performance**
- **Revenue**

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12. Benefits

- **Easy courier tracking**
- **Less paperwork**
- **Centralized information**
- **Faster data management**
- **Reduced manual errors**
- **Easy customer management**
- **Automated delivery updates**
- **Easy report generation**
- **Better courier monitoring**

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13. Future Enhancement

In future, the system can include:

- **GPS live tracking**
- **SMS notification**

  Email notification

  Online payment

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- **Mobile application**
- **QR code tracking**
- **Customer feedback**
- **AI-based delivery prediction**

Brainstorming & Ideation Phase

The brainstorming and ideation phase is the initial stage of the Courier Management System project. In this phase, different ideas are discussed to identify the problems faced in traditional courier management and to find suitable solutions using Salesforce.

The main ideas identified for the Courier Management System are:

- **Managing customer details in a centralized system.**
- **Creating and managing courier bookings.**
- **Maintaining sender and receiver information.**
- **Recording parcel details such as type and weight.**
- **Tracking courier delivery status.**
- **Assigning couriers to delivery agents.**
- **Managing courier charges and payment details.**
- **Automating delivery status updates using Salesforce Flow.**
- **Creating reports to monitor courier activities.**
- **Developing dashboards to display courier statistics.**

Based on these ideas, the Courier Management System was selected as the project concept. The proposed system aims to make courier operations simple, organized, and efficient by using Salesforce features.

Requirement Analysis Phase

The Requirement Analysis Phase is an important stage in the development of the Courier Management System. In this phase, the requirements of the courier company, customers, delivery agents, and administrators are identified and analyzed.

The main requirements identified for the Courier Management System are:

Functional Requirements

- **Customer registration and management.**
- **Courier booking and management.**
- **Sender and receiver details management.**
- **Parcel details such as type, weight, and destination.**
- **Courier delivery status tracking.**
- **Delivery agent assignment and management.**
- **Payment and courier charge management.**
- **Automatic status updates using Salesforce Flow.**
- **Report generation for courier activities.**
- **Dashboard for monitoring courier operations.**

Non-Functional Requirements

- **Security: Protect customer and courier information.**
- **Performance: Provide quick access to courier records.**
- **Reliability: Maintain accurate and consistent data.**
- **Usability: Provide a simple and user-friendly interface.**
- **Scalability: Support increasing numbers of customers and courier records.**

Based on these requirements, the Courier Management System can be developed using Salesforce to provide a centralized, secure, and efficient platform for managing courier operations.

Project Design Phase

The Project Design Phase defines the structure and functionality of the Courier Management System. In this phase, the system modules, Salesforce objects, fields, relationships, automation, reports, and dashboard are designed according to the requirements identified in the previous phase.

The main design components of the Courier Management System are:

1. Customer Management

Stores customer information such as name, phone number, email, and address.

2. Courier Management

Maintains courier booking details, sender and receiver information, source, destination, parcel type, weight, and delivery status.

3. Delivery Management

Manages delivery agents and assigns couriers for delivery.

4. Payment Management

Stores courier charges, payment method, payment date, and payment status.

5. Courier Tracking

Tracks the courier through different stages such as Booked, Picked Up, In Transit, Out for Delivery, and Delivered.

6. Salesforce Automation

Salesforce Flow is designed to automate status updates, delivery information, and notifications.

7. Reports and Dashboard

Reports and dashboards are designed to display total bookings, delivered couriers, pending deliveries, courier status, payments, and delivery performance.

The overall design provides a centralized and user-friendly system for managing the complete courier delivery process using Salesforce.

Project Planning Phase

The Project Planning Phase focuses on organizing and scheduling the development of the Courier Management System. In this phase, the project activities, resources, responsibilities, and development stages are planned to ensure the successful completion of the project.

The main activities planned for the Courier Management System are:

1. Requirement Planning

Identify the requirements of customers, administrators, and delivery agents.

2. System Planning

Plan the Salesforce objects, fields, relationships, and application structure.

3. Module Planning

Plan the major modules such as Customer Management, Courier Booking, Delivery Management, Payment Management, and Courier Tracking.

4. Salesforce Development Planning

Plan the creation of custom objects, fields, validation rules, Salesforce Flows, reports, and dashboards.

5. Data Planning

Plan how customer, courier, parcel, delivery agent, and payment information will be stored and managed.

6. Testing Planning

Plan testing activities to verify that courier booking, tracking, payment, and delivery functions work correctly.

7. Documentation Planning

Plan the preparation of project documentation, screenshots, system design, testing results, and final project report.

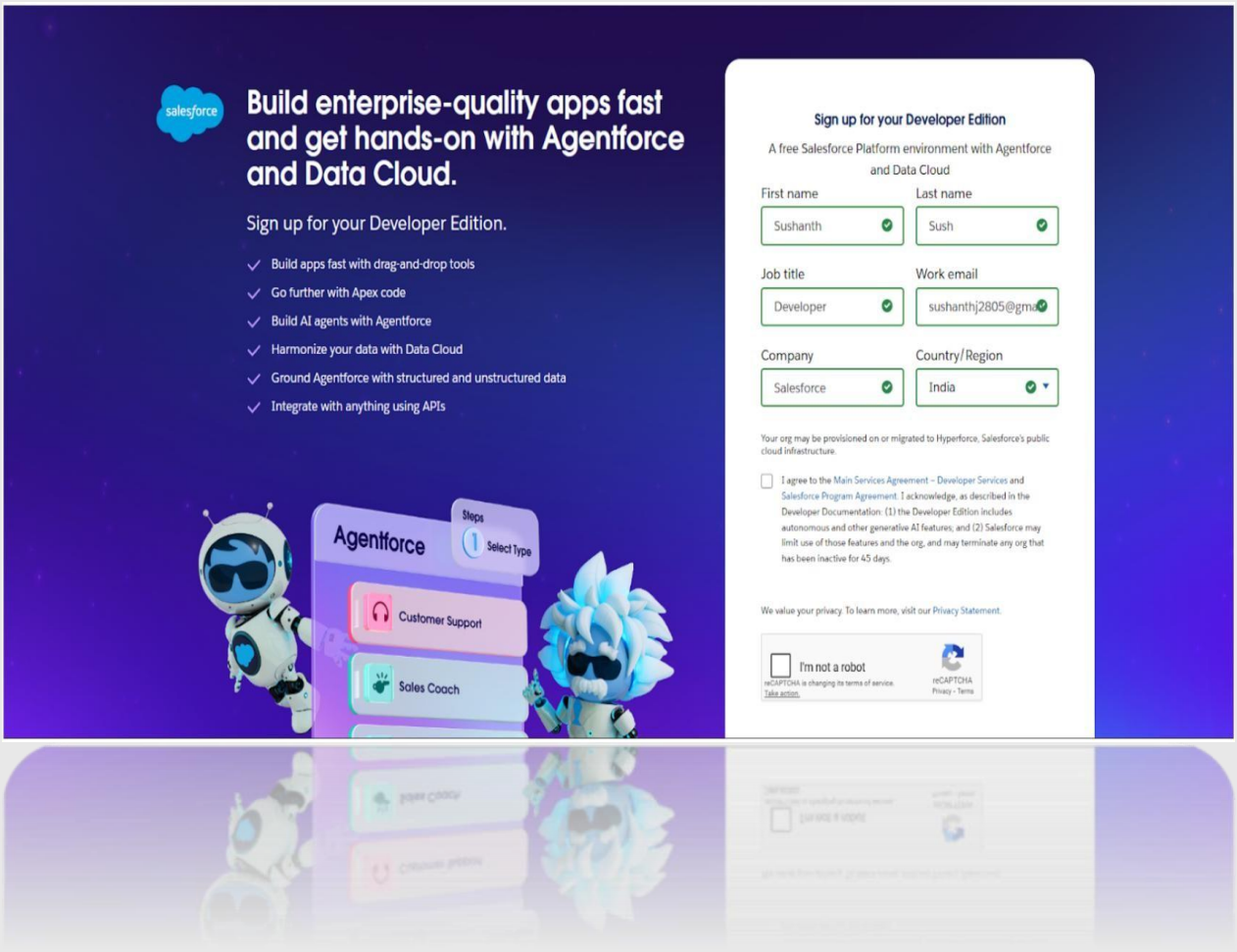
8. Project Schedule

The project is planned in different stages:

Requirement Analysis → System Design → Salesforce Development → Testing → Documentation → Demonstration

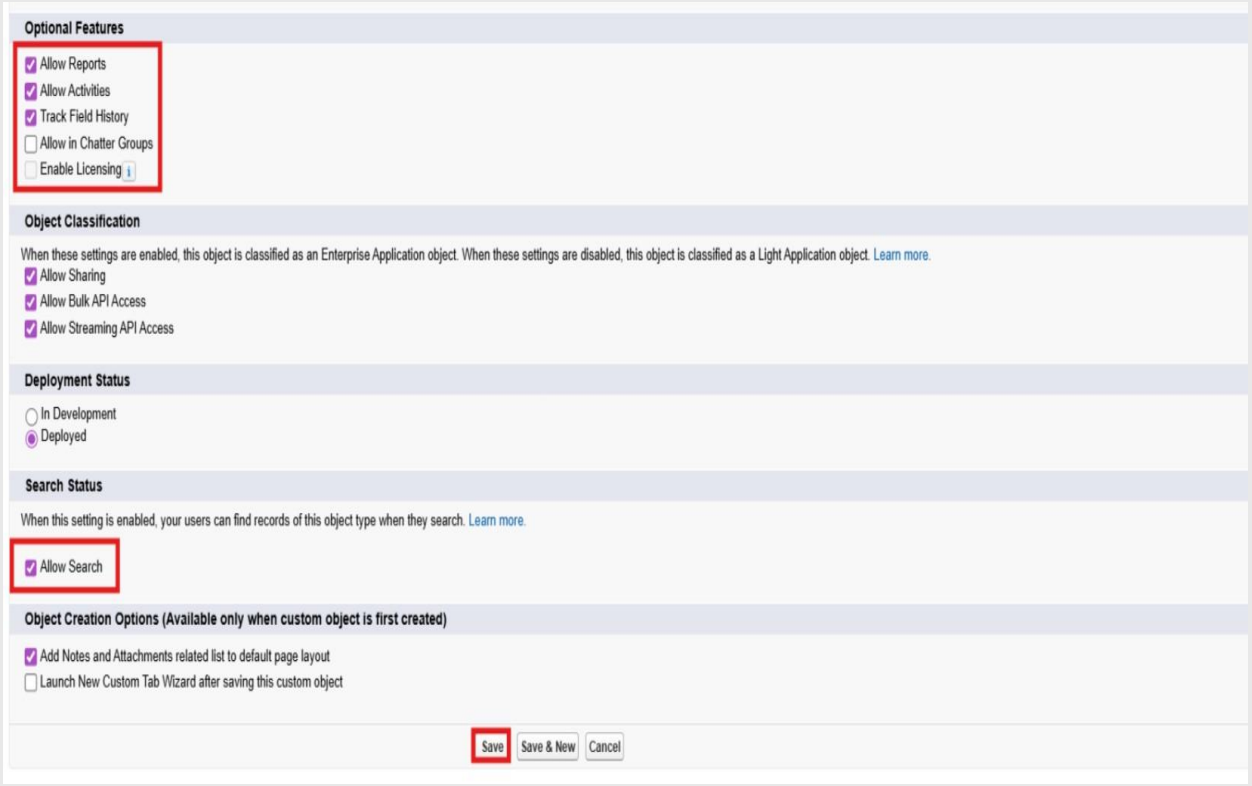
This planning helps the team complete the Courier Management System in an organized manner and ensures that all important project activities are completed successfully.

Project Development Phase

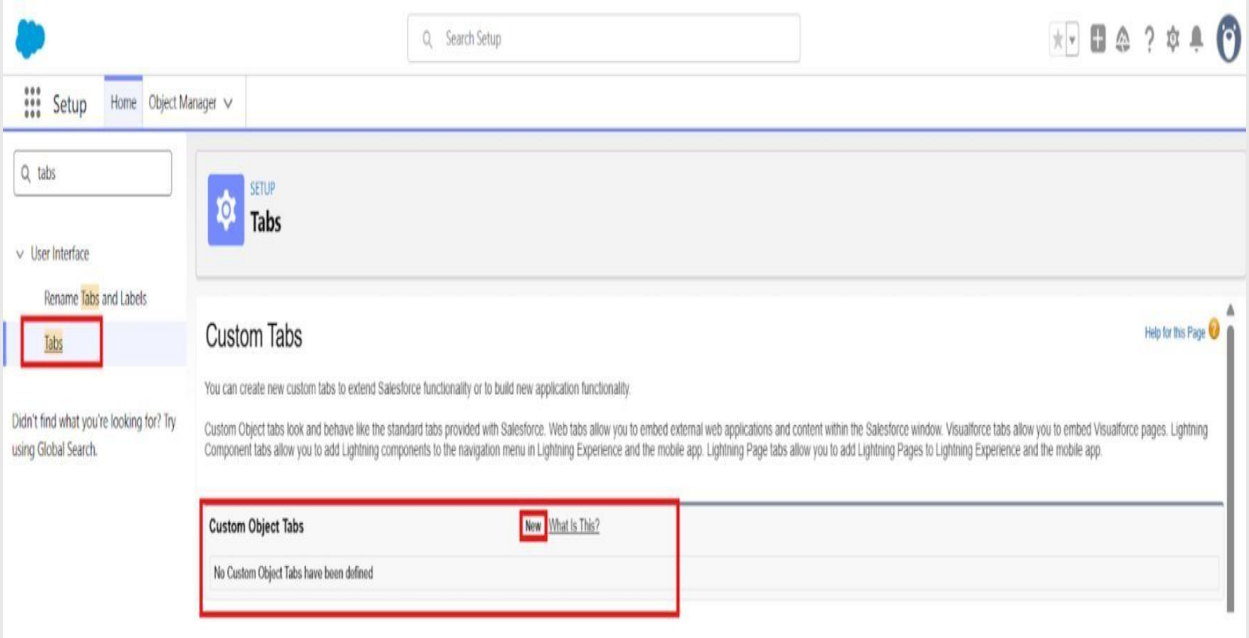


Project Testing Phase

Click Setup → Object Manager → Create → Custom Object



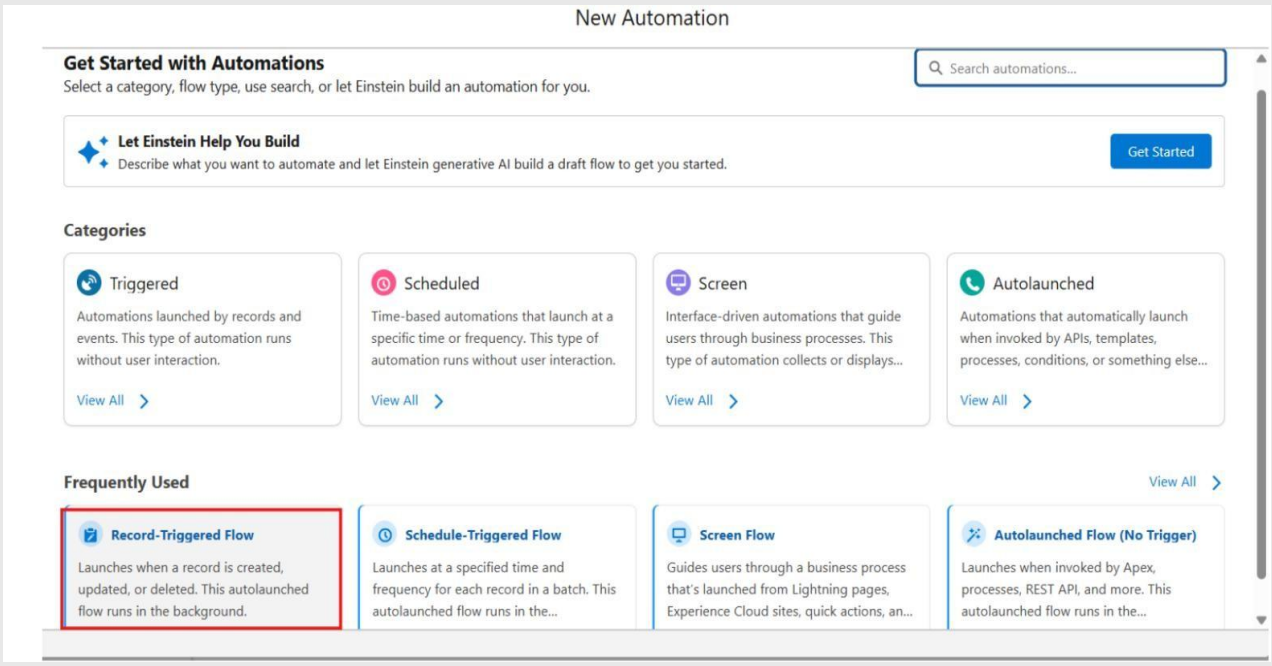
Creating a Tab for Customer Object



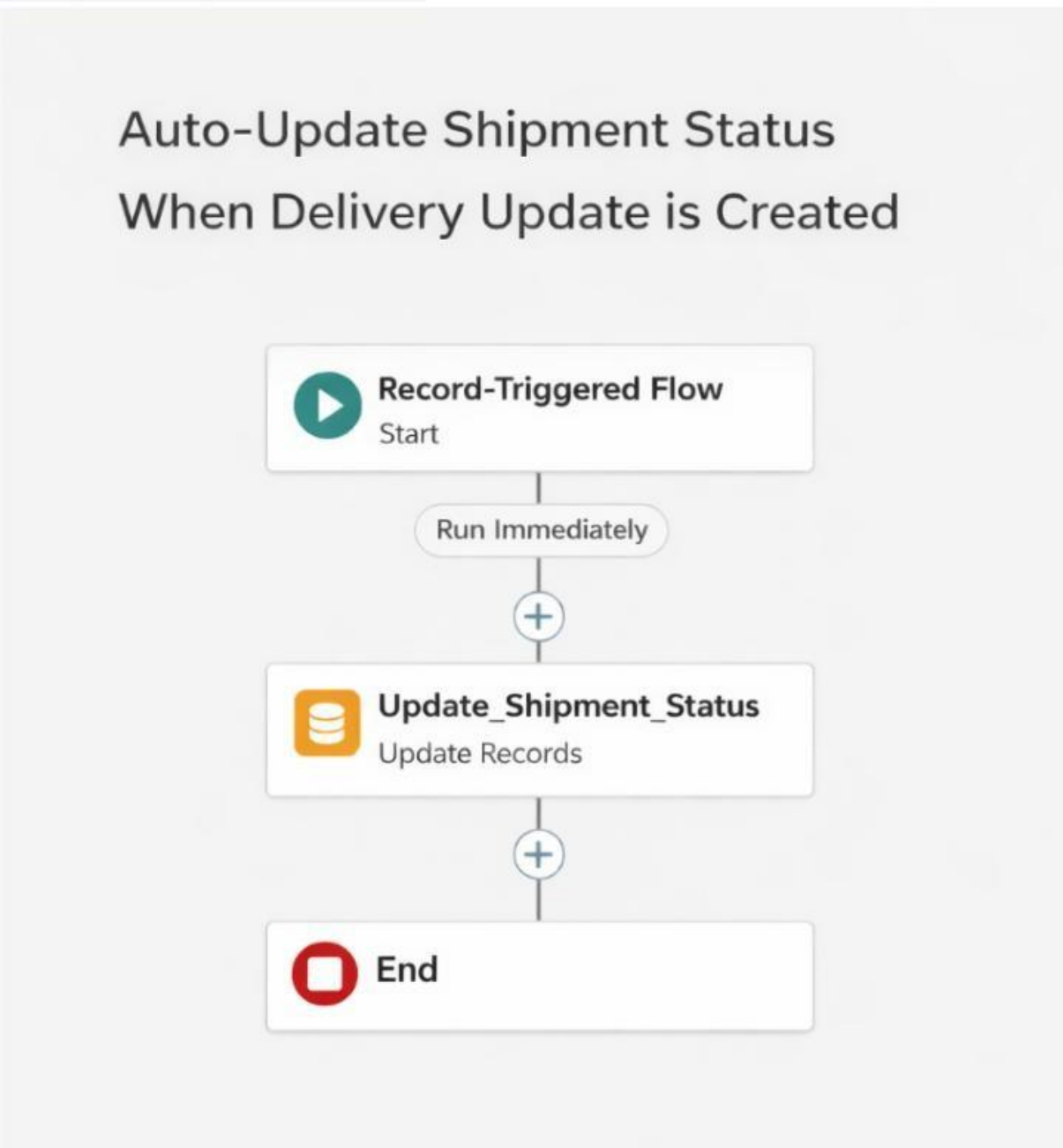
Creation of fields for the Customer Object

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of customer
Email	Email__c	Email	Yes	Customer Email
Phone	Phone__c	Phone	Yes	Customer contect number
Address	Address__c	Text Area	Yes	Customer address
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer City
State	State__c	Text	No	Customer State
Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

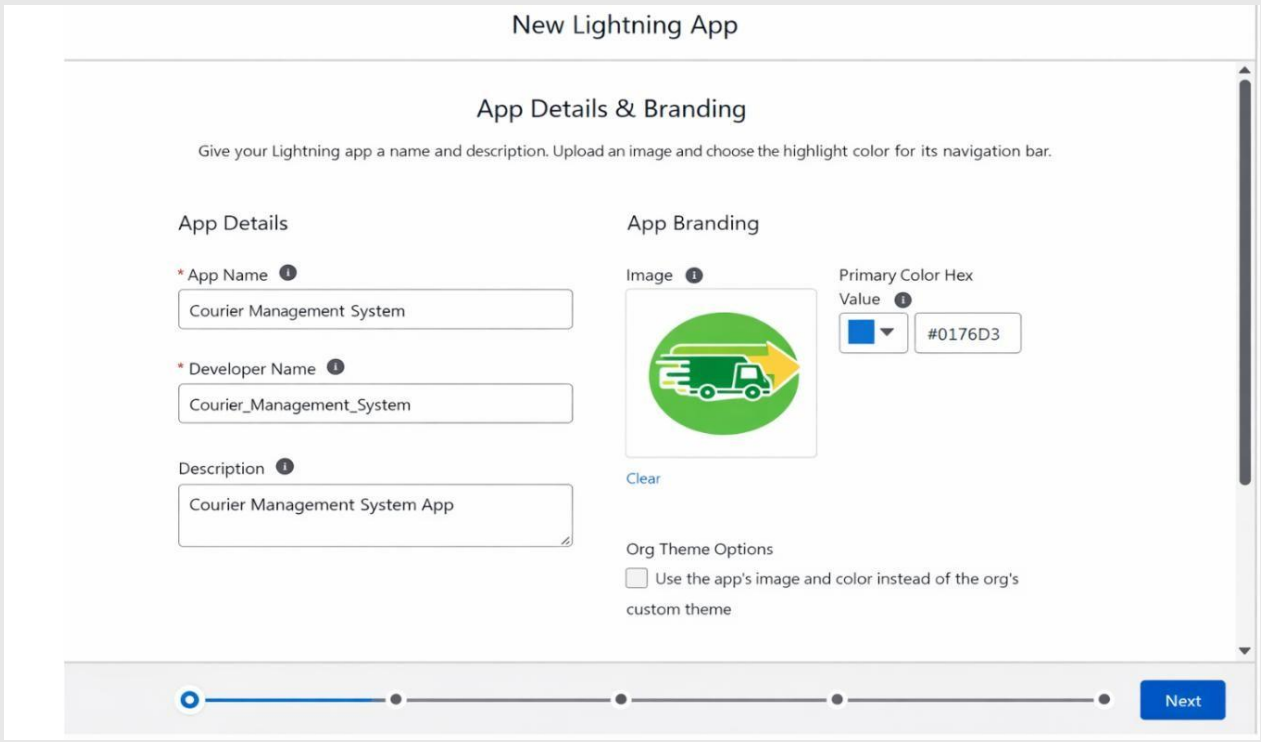
Create the Record-Triggered Flow



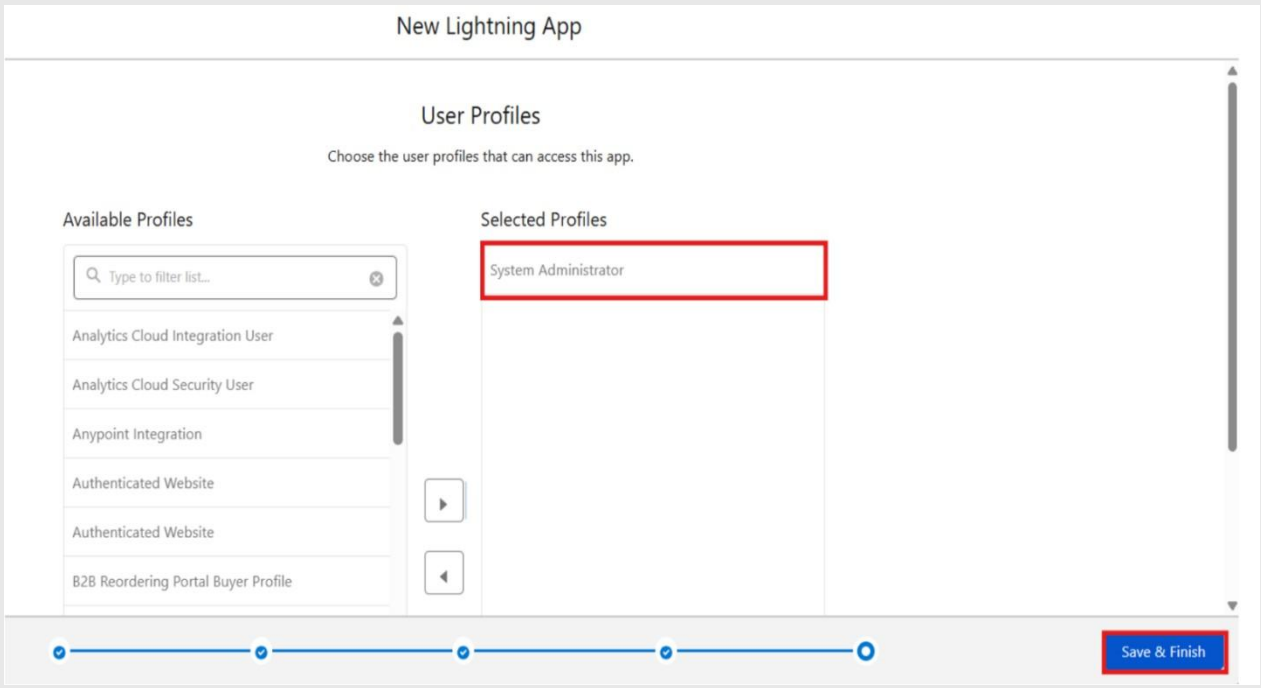
Final Flow Layout:



Create the Courier Management System App



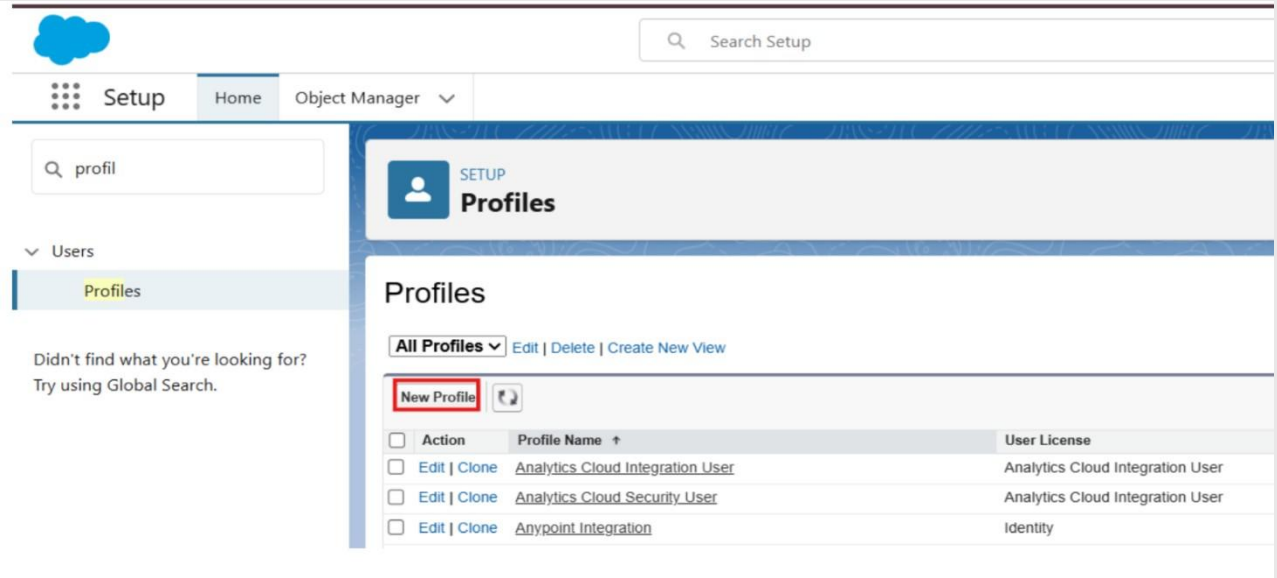
Create a Lightning App



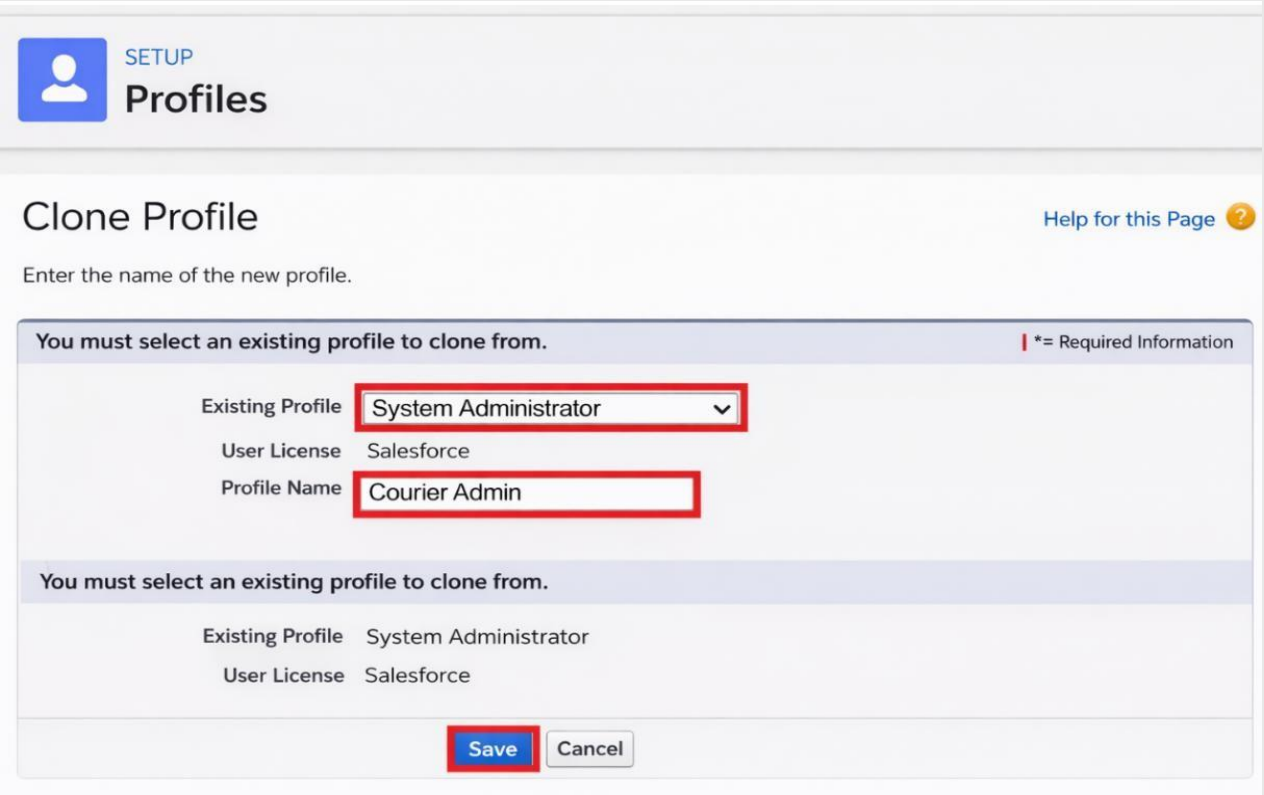
Project Documentation Phase

Create a Custom Profile – Courier Admin

1.



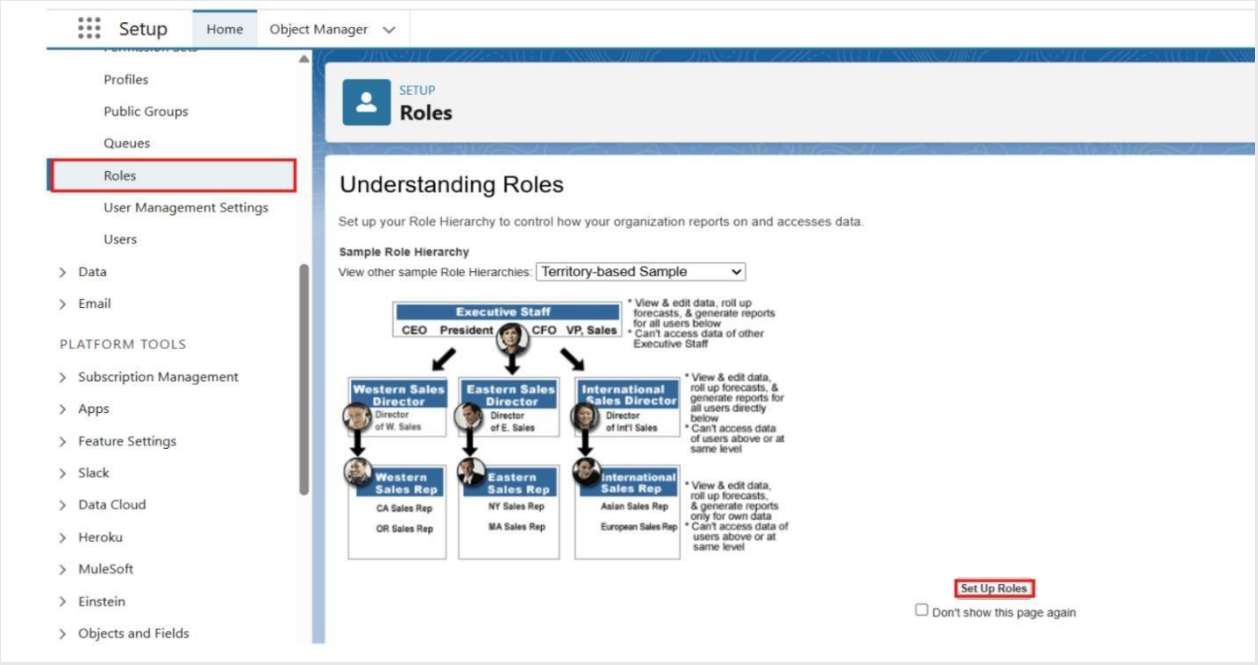
2.



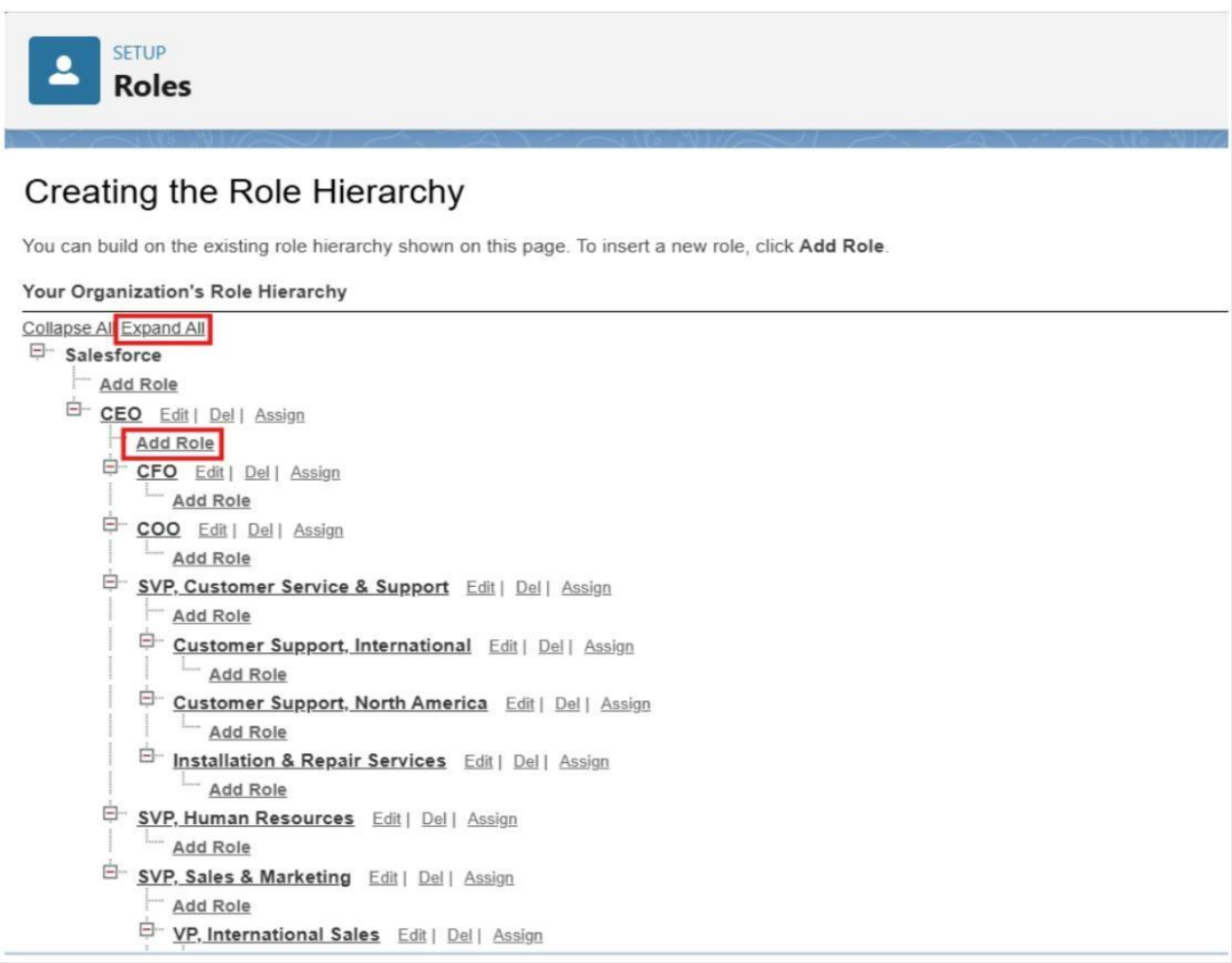
Project Demonstration Phase

Creation of Courier Admin Role

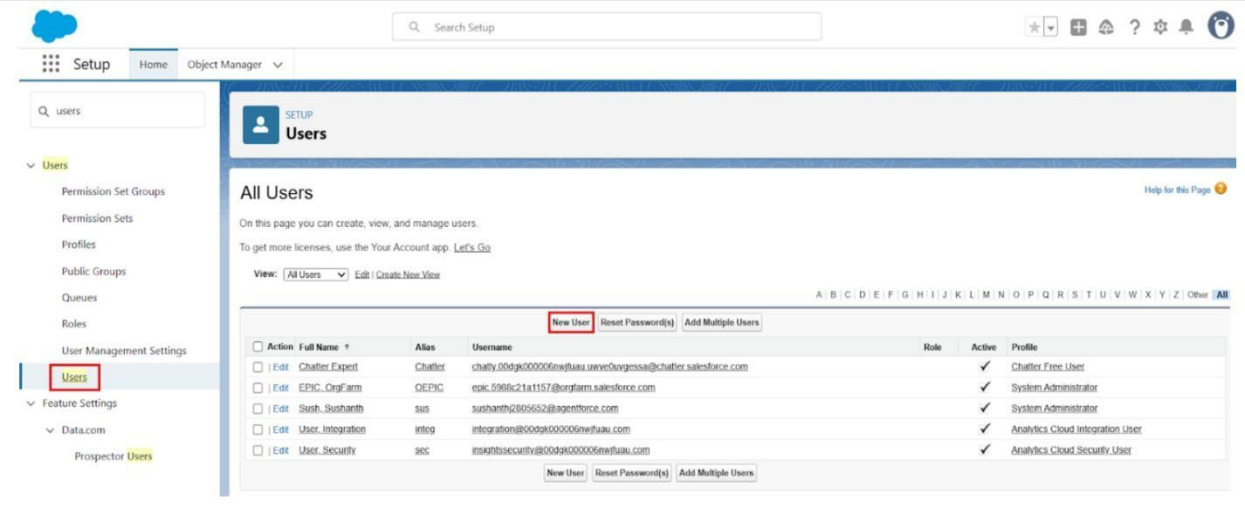
1.



2.



To Create a User



Save & Activate Dashboard



Conclusion

The Courier Management System provides an efficient and organized solution for managing courier operations. The system helps to manage customer details, courier bookings, parcel information, delivery agents, payments, and delivery status in a centralized platform.

By using Salesforce features such as custom objects, fields, relationships, automation, reports, and dashboards, the system reduces manual work and improves data management. It also helps users track courier deliveries and monitor courier activities easily. Overall, the Courier Management System improves the efficiency, accuracy, and reliability of courier

management and provides a user-friendly platform for handling the complete courier delivery process.